

Appx_18X — Internal Measure & Normalization Closure

Status: TRAILHEAD

Depends on: Appx_10 (Bounds), Appx_18 (7D Action & Metric), Appx_21 (GR Recovery Channel)

Enables: AoC_06 (Gravity Node)

§0 — Scope, Imported Assumptions, and Kill Conditions

This node closes the internal measure λ and determines whether κ_7 is an independent parameter or collapses into a ratio constraint.

It does not derive compactness of the internal sector, nor does it derive KK truncation validity. These are imported assumptions.

Imported Assumptions

1. Internal Compactness (if A1 route used).

Compactness of the internal regulator sector is inherited from Appx_18. It is not derived here.

2. KK Truncation Validity.

The $7D \rightarrow 4D$ reduction assumes:

- (i) The retained 4D graviton mode has no internal-coordinate dependence.
- (ii) Excited KK modes do not backreact onto the zero mode at leading order.
- (iii) Internal curvature scales remain separated from external curvature scales.

Violation of these invalidates the reduction.

3. Static Moduli (if assumed).

Unless otherwise stated, ζ , ω , ξ are treated as static in §1.3. Dynamic behavior is addressed in §2.

Expected Outcome

Given the current maturity of Appx_21, Case 2 (ratio-only determination) is the expected provisional outcome. Full independence of κ_7 requires further upstream development.

Kill Switches

This node fails if:

- Compactness cannot be justified upstream (A1 invalid).
- Integrability requires free decay parameters (A2 invalid).
- $\dot{G}/G$ violates empirical bound.
- KK normalization and GR recovery normalization disagree.
- λ depends on coordinate conventions.
- κ_7 is post-fit after λ evaluation.

§1 — Compact Internal Sector (Route A1)

1.1 Internal Coordinates

Define regulator coordinates:

$$y_a \sim y_a + L_a, \quad a \in \{\zeta, \omega, \xi\}$$

with strictly positive periods L_a .

Metric ansatz (product form):

$$h_{ab} = \text{diag}(\zeta^2, \omega^2, \xi^2)$$

ζ, ω, ξ are regulator moduli.

1.2 Moduli Classification

Static Case.

ζ, ω, ξ constant.

Dynamic Case.

ζ, ω, ξ spacetime-dependent fields.

1.3 λ Computation (Static Case Only)

$$\lambda = \int_{\mathcal{I}_3} \sqrt{\det h} d^3y = \int_0^{L_\zeta} \int_0^{L_\omega} \int_0^{L_\xi} (\zeta \omega \xi) dy_\zeta dy_\omega dy_\xi$$

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$$\lambda = (\zeta \omega \xi) L_\zeta L_\omega L_\xi$$

This expression holds under the Static Case (§1.2).

Under the Dynamic Case, λ becomes spacetime-dependent and is treated in §2.

1.4 Period Classification

Two possibilities:

1. Physics-fixed periods.

L_a derived from topology or regulator bounds.

2. Gauge/conventional periods.

Rescaling L_a leaves all physical predictions invariant.

Constraint:

If rescaling L_a changes α^G while leaving upstream invariants unchanged, then L_a are not gauge and must be derivable.

Failure to fix periods non-arbitrarily invalidates A1.

§2 — Dynamics Constraint

Derived from:

$$G_4 = \frac{\kappa_7}{8\pi\lambda}$$

Log-derivative:

$$\frac{\dot{G}_4}{G_4} = \frac{\dot{\kappa}_7}{\kappa_7} - \frac{\dot{\lambda}}{\lambda}$$

2.1 Static Case

$$\dot{\lambda} = 0 \Rightarrow \frac{\dot{G}_4}{G_4} = \frac{\dot{\kappa}_7}{\kappa_7}$$

If κ_7 fixed $\rightarrow \dot{G}_4 = 0$

2.2 Dynamic Case

$$\frac{\dot{\lambda}}{\lambda} = \frac{\dot{\zeta}}{\zeta} + \frac{\dot{\omega}}{\omega} + \frac{\dot{\xi}}{\xi} + \sum_a \frac{\dot{L}_a}{L_a}$$

2.3 Empirical Bound

Lunar laser ranging constraint:

$$\left| \frac{\dot{G}}{G} \right| \lesssim 10^{-13} \text{ yr}^{-1}$$

If dynamic moduli imply violation $\rightarrow$ node FROZEN.

§3 — Non-Compact Fallback (Route A2)

Activated only if compactness cannot be justified.

Define:

$$h_{ab}(y) = \text{diag}(f_\zeta(y)^2, f_\omega(y)^2, f_\xi(y)^2)$$

$$\sqrt{\det h} = f_\zeta f_\omega f_\xi$$

3.1 Integrability Theorem (Exponential)

If

$$f_a(y) \sim e^{-|y|/L_a}$$

then

$$\lambda = \int_{\mathbb{R}^3} \sqrt{\det h} d^3y < \infty$$

3.2 Integrability Theorem (Power-Law)

If

$$f_a(y) \sim |y|^{-p}$$

then integrable on $\mathbb{R}^3$ iff $p > 3$.

3.3 Derived Decay Obligation

Decay functions must be derived from regulator bounds (Appx_10) or firewall suppression arguments.

If decay constants are freely chosen $\rightarrow$ A2 invalid.

3.4 Uniqueness Clause

If multiple admissible decay profiles consistent with upstream bounds yield distinct λ values, the result must be expressed as a bounded interval rather than a point value.
If neither uniqueness nor bounded range can be established $\rightarrow$ A2 fails.

§4 — GR Recovery Consistency Lock (B3 Arbiter)

Purpose: determine whether κ_7 is independent.

4.1 KK Reduction Normalization

Under KK truncation assumptions:

$$\frac{1}{16\pi G_4} = \frac{\lambda}{2\kappa_7} G_4 = \frac{\kappa_7}{8\pi\lambda}$$

4.2 Independent GR Recovery (Appx_21 Channel)

Appx_21 recovers 4D Einstein equations with prefactor:

$$\frac{1}{16\pi G_4}$$

4.3 Equality Constraint

Both channels must yield identical normalization.

If not $\rightarrow$ node fails.

4.4 Outcome Classification

Case 1 — Ratio-only determination

Only κ_7/λ matters.

κ_7 becomes conventional once λ fixed.

Case 2 — Residual κ_7 freedom

κ_7 not fixed by ratio alone.

Operational consequence:

Appx_18X remains TRAILHEAD until:

- κ_7 independently derived (B2), or
- Framework demonstrates all predictions depend only on κ_7/λ .

Given current maturity of Appx_21, Case 2 is the expected provisional state.

κ_7 must never be post-fit after λ computation.

§5 — Closure Statement for AoC_06

If:

- λ finite (A1 or A2),
- $\dot{G}/G$ bound satisfied,
- KK and GR normalization agree,

then:

$$G_4 = \frac{\kappa_7}{8\pi\lambda}$$

is structurally admissible.

AoC_06 may substitute λ and κ_7/λ without introducing tunable normalization freedom.

This version incorporates:

- ✓ Renamed Static/Dynamic
- ✓ Boxed λ qualifier
- ✓ Explicit KK truncation assumption
- ✓ Compactness marked as imported
- ✓ A2 flagged as structural template
- ✓ Uniqueness clause added
- ✓ Case 2 operational status clarified
- ✓ Expected provisional outcome acknowledged

No expansion beyond what was requested.

Architecture preserved.